(For students admitted in 2025-26 under the 4-year degree)

BEng in Microelectronics and Integrated Circuits

In addition to the requirements of their major programs, students are required to complete the University requirements for graduation. For details please refer to the respective section on this website.

Students may use no more than 9 credits earned from courses offered in self-paced online delivery mode to satisfy the graduation requirements of a degree program. This 9-credit limit does not apply to credits obtained through the credit transfer procedures of the University.

For students graduating with an additional major, they must take all the requirements specified for that major, within which they must complete at least 20 single-counted credits. These 20 credits cannot be used to fulfill any other requirements for graduation except for the 120-credit degree requirement.

Under the new 30-credit Common Core Program which is applicable to students admitted to the University in 2022-23 and thereafter, courses that have been counted towards Major Requirements are not allowed to be reused for fulfilment of the University Common Core Requirements. Students should look up the details of the Common Core Program including the general and School-/program-specific distributional requirements posted on the Common Core website where the link to it is available on this website.

Major Requirements

Engineering Fundamental Course(s)

			Credit(s) attained
ELEC/MATH		Note: ELEC 2600 OR MATH 2011 OR MATH 2350 (2 courses out of 3)	6-7
ELEC	2600	Probability and Random Processes in Engineering	4
MATH	2011	Introduction to Multivariable Calculus	3
MATH	2350	Applied Linear Algebra and Differential Equations	3
COMP	1023	Introduction to Python Programming	3
COMP		Note: COMP 2011 OR COMP 2012H	4-5
COMP	2011	Programming with C++	4
COMP	2012H	Honors Object-Oriented Programming and Data Structures	5
MATH		Note: [(MATH 1013 OR MATH 1023) AND (MATH 1014 OR MATH 1024)] OR [MATH 1020]	4-6
MATH	1013	Calculus I	3
MATH	1014	Calculus II	3
MATH	1020	Accelerated Calculus	4
MATH	1023	Honors Calculus I	3
MATH	1024	Honors Calculus II	3
PHYS		Note: PHYS 1112 OR PHYS 1312	3
PHYS	1112	General Physics I with Calculus	3
PHYS	1312	Honors General Physics I	3
PHYS		Note: PHYS 1114 OR PHYS 1314	3
PHYS	1114	General Physics II	3
PHYS	1314	Honors General Physics II	3

Required Course(s)

			Credit(s) attained
ELEC	1100	Introduction to Electro-Robot Design	4
ELEC	1300	Introduction to Microelectronics and Integrated Circuits	4
ELEC	1910	Academic and Professional Development I	0
ELEC	2100	Signals and Systems	4
ELEC	2350	Introduction to Computer Organization and Design	4
ELEC	2400	Electronic Circuits	4
ELEC	2910	Academic and Professional Development II	0
ELEC	2992	Industrial Experience (Microelectronics and Integrated Circuits)	0
ELEC	3400	Introduction to Integrated Circuits and Systems	4
ELEC	3500	Integrated Circuit Devices	4
ELEC	3910	Academic and Professional Development III	0
ELEC		Note: [(ELEC 4920 OR ELEC 4930) AND (ELEC 4900 OR ELEC 4901 OR ELEC 4910)] OR [(ELEC 4930 AND ELEC 4931 AND ELEC 4932)] (Students taking the Research Option must take ELEC 4901)	9
ELEC	4900	Final Year Design Project	6
ELEC	4901	Final Year Thesis	6
ELEC	4910	Co-op Program	6
ELEC	4920	Microelectronics Project	3
ELEC	4930	Integrated Circuit Design Project I	3
ELEC	4931	Integrated Circuit Design Project II	3
ELEC	4932	Integrated Circuit Design Project III	3

Elective(s)

			Minimum credit(s) required
ELEC		ELEC Electives (Courses from the specified elective list, of which at least 2 courses must be at 4000-level.)	13
ELEC	3300	Introduction to Embedded Systems	4
ELEC	3310	Digital Fundamentals and System Design	4
ELEC	3410	CMOS VLSI Design	3
ELEC	3600	Electromagnetics: From Wireless to Photonic Applications	4
ELEC	4010P	Quantum Mechanics for Electronic Engineers	3
ELEC	4010R	Compound Semiconductor Device Physics and Technologies	3
ELEC	4010S	Microwave Theory and Applications	4
ELEC	4320	FPGA-based Design: From Theory to Practice	3
ELEC	4420	Analogue Integrated Circuits Design and Analysis	4
ELEC	4430	Integrated Power Electronics	3
ELEC	4510	Semiconductor Physics for Solid-State Electronics	3
ELEC	4520	Integrated Circuit Fabrication Technology	3
ELEC	4530	Fundamentals of Photovoltaic and Renewable Energy	3

ELEC	4610	Engineering Optics	4
ELEC	4620	Photonics and Optical Communications	4
ELEC	4810	Introduction to Biosensors and Bioinstrumentation	4

Students may opt to graduate with or without an option. Students who take an option MUST complete all requirements specified in addition to the major requirements.

Option(s)

Research Option

Students in the Research Option should also take ELEC 4901 as specified in the major requirements.

Required Course(s)

			Credit(s) attained
ELEC	5900	Modern Engineering Research Methodologies	3

Elective Course(s)

			Minimum credit(s) required
		Advanced Elective Courses approved by advisor (at least one UROP course taken prior to the commencement of Final Year Thesis, and one PG-level course)	6
UROP	1000	Undergraduate Research Opportunities	0
UROP	1100	Undergraduate Research Opportunities Series 1	1
UROP	2100	Undergraduate Research Opportunities Series 2	1
UROP	3100	Undergraduate Research Opportunities Series 3	1
UROP	4100	Undergraduate Research Opportunities Series 4	1